

4.4 DEPOSIT CREATION

MARKING SCHEME

1990/CE/II/45 C	1996/CE/II/36 B	2001/CE/II/26 C	2005/CE/II/37 A (57%)	2010/CE/II/35 B (68%)
1991/CE/II/31 B	1997/CE/II/33 B	2002/CE/II/35 C (84%)	2005/CE/II/50 B (46%)	2010/CE/II/36 C (43%)
1991/CE/II/32 B	1997/CE/II/34 A	2002/CE/II/36 D (56%)	2006/CE/II/36 D (61%)	2012/DSE/I/27 C (71%)
1992/CE/II/34 B	1998/CE/II/33 A	2002/CE/II/37 C (59%)	2006/CE/II/37 B (57%)	2013/DSE/I/26 C (58%)
1992/CE/II/35 D	1998/CE/II/34 D	2002/CE/II/38 A (25%)	2006/CE/II/38 C (37%)	2014/DSE/I/30 D (56%)
1992/CE/II/32 C	1998/CE/II/52 C	2003/CE/II/34 D (17%)	2007/CE/II/36 D (49%)	2016/DSE/I/31 D (65%)
1994/CE/II/40 A	1999/CE/II/36 B	2003/CE/II/35 B (24%)	2008/CE/II/38 B (39%)	2016/DSE/I/33 A (deleted)
1994/CE/II/42 D	1999/CE/II/38 A	2003/CE/II/36 C (16%)	2008/CE/II/39 D (60%)	2017/DSE/I/31 B (36%)
1995/CE/II/42 B	2000/CE/II/29 C	2004/CE/II/38 A (73%)	2009/CE/II/38 A (44%)	2019/DSE/I/28 D
1996/CE/II/35 C	2000/CE/II/30 B	2004/CE/II/39 B (34%)	2009/CE/II/39 D (37%)	2020/DSE/I/33 B
2022/DSE/I/30 D	2022/DSE/I/32 C	2023/DSE/I/34 C	2024/DSE/II/34 B	2025/DSE/II/34 D

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/CE/2/(d)(ii)

(I) $\$10\,000 \times (1 - 25\%)$ (2)

= \$7 500 (1)

Possible way: to recall loans / sell investment assets (2)

(II) $\$10\,000 \times (1 / 25\%)$ (2)

= \$40 000 (1)

1991/CE/I/5(b)

(i) $\$200 / \$800 = 25\%$ (2)

(ii) Actual reserves – (Deposits $\times$ Required reserve ratio)
 $\$50 - (\$100 \times 25\%) = \$25$ (2)

(iii) Excess reserves $\times$ Banking multiplier
 $\$25 \times (1 / 25\%) = \100 (3)

Assumptions:

- no cash drain (2)

- people are willing to borrow (2)

1992/CE/I/3(b)(ii)

- 100% reserves required by the government
 - the bank keeps all the \$1000 as reserves
 - no one borrows from the bank
 - any cash that leaks out of the bank is not redeposited with the banking system
- (Mere listing of points without elaboration - max. 2 marks.)

(3@, max: 6)

[Mark the **FIRST TWO** points only.]

1993/CE/I/5(c)(ii)

$$\begin{aligned} & \$3 \text{ million} \times (1 / 25\%) \\ & = \$12 \text{ million} \end{aligned}$$

(2)

(1)

1994/CE/I/5

$$\begin{aligned} & \text{Maximum amount of deposits to be created} = \$100 \times (1 / 20\%) \\ & = \$500 \end{aligned}$$

(3)

(1)

1995/CE/I/10(a)(iii)

Brief description of the changing process in a correct sequence:

Initial withdrawal of \$M $\Rightarrow$

(Step 1:) Banks lose reserves (i.e. actual reserves < required reserves)

(2)

(Step 2:) Banks call back loans / sell bank assets to the public

(2)

(Step 3:) Further withdrawal of bank deposits to repay the loans / to buy bank assets
The process of bank deposit contraction goes on and on

(2)

(1)

(Step 4:) Ultimately, total bank deposits contract
by $\$M \times (1 / 20\%) = \$5M$

(1)

(2)

(max: 8)

1996/CE/I/11(c)

- (i) No change in total bank deposits, because
it is only a switch from one's bank deposit to another's.

(1)

OR

there may be some deposit contraction at first, but then it would be offset by the deposit expansion later on.

(2)

- (ii) Banks gain excess reserves

(1)

$\Rightarrow$ increase in total bank deposits

(1)

by an amount of $\$5mn \times (1 / 25\%) = \$20mn$

(2)

1997/CE/I/9(c)

(A logical sequence of the following is required.)

An injection of new money (e.g. the initial \$10 b.) to the local banks

(1)

$\Rightarrow$ banks gain excess reserves (\$R)

(1)

and will loan out the excess reserves

(1)

$\Rightarrow$ the loan will be redeposited into the banking system

(1)

But due to the following factor(s), the redeposit amount is smaller than SR:

- initially not all the new money (the \$10b.) are deposited to the local banks
- local banks hold excess reserves
- insufficient demand for loan
- cash drain

(2@, max: 4)

Elaboration on how any one of the above factors would lead to a total bank deposit amount smaller than the maximum amount

(2)

(max: 8)

1998/CE/1/9(a)

(Step 1:) After the bank accepts Mr Wong's \$1 m deposit, it will lend out / invest 75% or \$0.75 m in the first round (2)

(Step 2:) The \$0.75 m loan / investment will be redeposited into the banking system (2)

The process of (Step 1 - Step 2 - Step 1...) will go on and on until (2)

finally the total deposit created = initial deposit $\times$ (1 / reserve ratio)

= $\$1 \times (1 / 25\%) = \4 m (2)

1999/CE/1/6(a)

$\$10\,000 + [\$10\,000 \times (1 - 25\%)]$ (2)

= \$17 500 (2)

2000/CE/1/10(a)

Answer 1

(Assume that the public does not use the \$50 mn cash in hand to repay bank loans when the banks call back loans due to the withdrawals)

(i) $-\$10\text{ mn} \times [1 - (200 / 1\,000)]$ (2)

= $-\$50\text{ mn}$ (1)

(ii) Change in money supply

= Change in cash in hand + Change in bank deposits (1)

= $(+\$10\text{ mn}) + (-\$50\text{ mn})$ (2)

= $-\$40\text{ mn}$ / a contraction of \$40 mn (1)

Answer 2

(Assume that the public considers to make use of the \$50 mn cash in hand to repay bank loans when the banks call back loans due to the withdrawals)

(i) The max. change is the same as the one in Answer 1. However, the actual change depends on how much of the \$50 mn cash in hand is used to repay the bank loans. The min. change is $-\$10\text{ mn}$. (Max: 3)

(ii) Use the same formula in Answer 1. The change in bank deposits varies from a max. decrease of \$50 mn to a min. decrease of \$10 mn, depending on how much of the \$50 mn cash in hand is used to repay the bank loans. The change in cash in hand varies from a max. to $+\$10\text{ mn}$ to a min. of $+\$2\text{ mn}$. Therefore, the final numerical answer varies from $-\$40\text{ mn}$ to $-\$8\text{ mn}$. (Max: 4)

2001/CE/1/10(b)

(i) Actual reserve ratio = $250 / 1\,000$ (1)

= 0.25 (1)

Excess reserve = $(0.25 - 0.2) \times \$1\,000$ (1)

= \$50 mn (1)

(ii) The withdrawal reduces the amount of reserves to \$210 mn. (1)

The new amount of reserves can back up $\$210\text{ mn} \times (1 / 20\%)$ (2)

= \$1 050 mn of deposits (1)

2002/CE/1/10(a)

(i) The required reserve ratio = $(20 / 100) = 0.2$ or multiplier = 5 (1)

Max. amount of deposits created = $\$10\text{ m} \times 5 = \50m (1)

Max. amount of money supply created = Max. amount of deposits created - decrease in cash in public circulation (1)

= $(\$50\text{m} - \$10\text{m}) = \$40\text{ m}$ (2)

(ii) - there is cash leakage

- the banks decide to hold excess reserves

- insufficient demand for loans

(2@, max: 4)

[Mark the **FIRST TWO** points only.]

2003/CE/I/11

- (b) (ii) Some banks cannot lend out their excess reserve as the demand for bank loans ↓
Step 1: loans that can be made by banks would decrease (1)
Step 2: money re-deposited into the banking system would decrease (1)
Correct sequence : Step 1 → Step 2 (1)
Therefore, deposit creation ability ↓ (1)
- (c) Legal reserve ratio = 0.2 (1)
Max. deposit = $\$250 \text{ m} \times (1 / 20\%)$ (1)
= $\$1\,250 \text{ m}$ (2)

2004/CE/I/6(b)

- Initial withdrawal of deposits,
⇒ banks lose reserves / insufficient reserves / actual reserves smaller than required reserves (1)
⇒ banks have to call back loans / sell assets to the public (1)
⇒ further withdrawals of deposits (1)
The process goes on and on. (1)

2005/CE/I/5

- (a) The amount of excess reserve
= $\$[120 - (1\,000 \times 10\%)] \text{ m} = \20 m (1 + 1)
- (b) Maximum possible amount of bank deposits
= $\$120 \text{ m} \times (1 / 10\%) = \$1\,200 \text{ m}$ (1 + 1)

2006/CE/I/6

- (a) 10% (1)
- (b) Yes because (1)
when total deposits reduce to $\$1\,150 (= \$1\,500 - \$350)$,
it only needs a reserve of $\$115 (= \$1\,150 \times 10\%)$ to back up the deposits.
Since the reserve assets only reduce to $\$150 (= \$500 - \$350)$, there is still an excess reserve of $\$35$. (2)

2007/CE/I/7

- (a) $\$200$ (1)
- (b) (i) $-\$100$ (1)
- (ii) $-\$30$ (2)

2008/CE/I/10

- (a) An increase in the reserve ratio
⇒ insufficient reserves / actual reserves smaller than required reserves (1)
⇒ banks have to call back loans / sell assets to the public (1)
⇒ withdrawals of deposits (1)
The process will go on and on (until the actual reserve is equal to the required reserve).* (1)
[*Remark: This point deserves one mark only when the sequence of the above steps is correct]

- (b) New required reserve ratio = $0.2 + 0.05 = 0.25$

Amount of bank deposits that \$1 000 million reserve can back up = $\$1\,000 \text{ million} \times (1 / 25\%)$ (1)
 = \$4 000 million (1)

OR

Change of bank deposits = $-\$250 \text{ million} \times (1 / 25\%)$ (1)
 = $-\$1\,000 \text{ million}$ (1)

Change in MS = Change in Cp + change in Dd / No change in Cp / MS = Cp + Dd (1)
 Change in MS = $-\$1\,000 \text{ million}$ (1)

(MS = money supply;

Cp = cash in the hands of the non-bank public; and

Dd = bank deposits.)

2009/CE/I/6

- (a) The minimum required reserve is $\$1\,000 \times 20\% = \200 (1)
 Excess reserve = \$100 (1)

- (b) Total amount of deposits at maximum = $\$300 \times (1 / 20\%)$ (1)
 = \$1 500 (1)

OR

Amount of extra deposits created = $\$100 \times (1 / 20\%)$ (1)
 Therefore, total amount of deposits at maximum = $\$1\,000 + \$500 = \$1\,500$ (1)

2010/CE/I/11(b)

- (i) Banking multiplier = $1 / 40\% = 2.5$ (1)

The max. amount of deposits that can be supported by \$2 000 million reserve at 40% reserve ratio

= $\$2\,000 \text{ mn.} \times (1 / 0.4)$ (1)
 = \$5 000 mn. (1)

Therefore, the max. change in bank deposits = $\$5\,000 \text{ mn.} - \$8\,000 \text{ mn.} = -\$3\,000 \text{ mn.}$ (1)

- (ii) An increase in the reserve ratio
 ⇒ the actual reserve is smaller than the desired reserve (1)
 ⇒ banks have to call back loans / sell assets to the public (1)
 ⇒ withdrawals of deposits (1)

The process will go on and on (until the actual reserve is equal to the desired reserve). * (1)

[*This 4th mark will be awarded only if the above three marks are already awarded.]

[#If the process starts with “withdrawals of deposits”, then award only a maximum of 2 marks]

1997/AL/II/9

- (a) reserve ratio = 0.25

- (b)

Assets (\$)		Liabilities (\$)	
Reserves	1 000	Deposits	5 500
Loans	4 500		

When the required reserve ratio is 10%, the maximum loan the banking system can extend is \$9 000. Since the maximum demand for loan is \$4 500, the loan extended will be \$4 500. Money supply is \$5 500.

- (c) A government can restrict the maximum amount a banking system can lend, or the maximum money supply. But the actual amount of lending or the actual money supply is also affected by the demand for loan.

1998/AL/II/9

- (a) Money supply = Deposits + amount of currency held by the public
= \$4 000 + \$500 = \$4 500

(b) (i)

Assets (\$)		Liabilities (\$)	
Reserves	1 100	Deposits	4 100
Loans	3 000		

$$\text{Money supply} = \$4\ 100 + \$400 = \$4\ 500$$

(ii)

Assets (\$)		Liabilities (\$)	
Reserves	1 100	Deposits	4 400
Loans	3 300		

$$\text{Money supply} = \$4\ 400 + \$400 = \$4\ 800$$

- (c) It will increase the money supply only if the banks can lend out some or all of their excess reserves.

2017/DSE/I/8

- (a) $\text{RRR} = (\$1\ 000 - \$250) / \$3\ 000 = 0.25$

(1)

- (b) The change in deposits = $\$250 \times (1 / 0.25) = \$1\ 000$

(2)

2019/DSE/II/7

- (a) **Excess reserves** = \$300 million – \$1000 million x 20% = \$100 million (1)

- (b) **Old money supply** = \$1000 million + \$200 million = \$1200 million (4)

$$\text{New money supply} = (\$300 \text{ million} + \$200 \text{ million}) \times 1/20\% + \$0 \text{ million} = \$2500 \text{ million}$$

$$\text{Maximum possible change in the money supply} = \$2500 \text{ million} - \$1200 \text{ million} = \$1300 \text{ million}$$

2021/DSE/II/6

6. (a) Required reserve ratio = $(600 - 300) / 1500 = 20\%$ (1)

- (b) New required reserve ratio = $20 - 5 = 15\%$ (3)

$$\text{New deposit} = 1500 + (600 - 0.15 \times 1500) / 0.15 = 4000 \text{ (\$million)}$$

$$\text{New money supply} = 4000 + 100 = 4100 \text{ (\$million)}^\dagger$$

In real-world practice of monetary policy, adjustment in the required reserve ratio (RRR) is understood as a change in percentage point or basis point (i.e. a level change in RRR) rather than a percentage change in RRR. The following answers are also accepted in case the candidate (mis)understood the 5% reduction in RRR as a percentage change rather than change in percentage point.⁵

$$\text{New required reserve ratio} = 20\% \times (1 - 5\%) = 19\%$$

$$\text{New deposit} = \$3157.89 \text{ million}$$

$$\text{New money supply} = \$3257.89 \text{ million}$$

2022/DSE/IB/8

(a) 25%.

- (b) Original money supply $= 4000 + 1000 = 5000$ (\$million)
- New deposit $= (1000 + 600) \times \frac{1}{25\%} = 6400$ (\$ million)
- New money supply $= 6400 + 400 = 6800$ (\$million)
- Change in Ms $= 6800 - 5000 = 1800$ (\$million)

2023/DSE/II/10i, ii

- (b) (i) Expected real rate of return $= 4.5\% - 4.5\% = 0\%$ (1)
Actual real rate of return $= 4\% - (-1\%) = 5\%$ (1)
- (ii) Gain, because (1)
the actual real rate of return is higher than the expected real rate of return. (1)

2024/DSE/II/8

- (a) The required reserve ratio $= [(\$600 \text{ million} + \$100 \text{ million}) / \$3\,500 \text{ million}] \times 100\%$ (1)
 $= 20\%$
- (b) The new required reserve ratio $= 20\% - 5\% = 15\%$ (1)
The maximum amount of deposits after deposit creation (1)
 $= \$600 \text{ million} / 15\% = \$4\,000 \text{ million}$
The maximum change in deposits (1)
 $= \$4\,000 \text{ million} - \$3\,500 \text{ million}$
 $= \$500 \text{ million}$